ANNEXURE III: VACUUM & CRYOGENIC SPECIFICATION SCHEDULE

Technical specifications for high-vacuum cryogenic probe station. Borderless continuation matrix.

Sl. No.	Subsystem / Parameter	Technical Tender Specification	Compliance (Yes/No)	Remarks / Justifications
1.0	Vacuum Chamber	Ultimate base pressure: <= 1.0 x 10^-7 mbar within 4 hours of turbo pumping;		
2.0	Turbo Pump	Pumping speed for N2: >= 600 L/s with integrated magnetic levitation bearing;		
3.0	Roughing Pump	Dry scroll vacuum backing pump with pumping speed of >= 15 m3/h;		
4.0	Vacuum Gauging	Wide range combination gauge (Pirani / Cold Cathode) from 1000 mbar to 10^-9 mbar;		
5.0	Cryocooler Unit	Two-stage closed cycle pulse tube cryocooler providing 1.0 W cooling power at 4.2 K;		
6.0	Vibration Damping	Bellows vibration isolation interface with residual vibration amplitude < 50 nm RMS;		
7.0	Temperature Controller	3-channel cryogenic temperature controller with 0.001 K sensor reading resolution;		
8.0	Sample Mount	Oxygen-free high conductivity (OFHC) copper sample stage with 50 mm diameter;		
9.0	Optical Viewport	Fused silica optical windows with anti-reflective coating for 400 nm to 1100 nm;		
10.0	Coaxial Cabling	Four semi-rigid coaxial lines rated for DC to 40 GHz with SMA connectors;		
11.0	Radiation Shield	Gold plated copper radiation shield minimizing thermal radiative heat transfer;		
12.0	System Warranty	Comprehensive 3-year on-site warranty covering cryocooler, pumps and controller;		